International Journal of Pediatric Otorhinolaryngology xxx (2015) xxx–xxx

Contents lists available at ScienceDirect

International Journal of Pediatric Otorhinolaryngology

journal homepage: www.elsevier.com/locate/ijporl

Paediatric tracheostomy—An 11 year experience at a Scottish paediatric tertiary referral centre

C.M. Douglas [a,*], J. Poole-Cowley [b], S. Morrissey [a], H. Kubba [a], W.A. Clement [a], D. Wynne [a]

[a] Department of Paediatric Otolaryngology, Royal Hospital for Sick Children, Dalnair St., Glasgow G3 8SJ, Scotland
[b] School of Medicine, University of Glasgow Medical School, Glasgow, Scotland

ARTICLE INFO

Article history:
Received 12 May 2015
Received in revised form 15 July 2015
Accepted 16 July 2015
Available online xxx

Keywords:
Paediatric
Tracheostomy
Paediatric airway

ABSTRACT

Aims: The aim of this paper was to review the indications, complications and outcomes for tracheostomy at a Scottish paediatric tertiary referral hospital.
Methods: All patients undergoing tracheostomy between January 2001 and September 2012 were identified. A retrospective case note analysis was performed.
Results: 111 tracheostomies were done in the study period. The mean number per year was 11 (3–12). Full data was available for 95 patients. There were 56 (59%) males and 39 (41%) females. Age at time of tracheostomy ranged from one day to 15 years, the mean age of tracheostomy insertion was 69 weeks. The majority of patients, 75 (79%), were under one year old when they had their tracheostomy. The most common indication was long-term ventilation (20%), followed by craniofacial abnormality causing airway obstruction (18%), followed by subglottic stenosis (14%). 37% of patients were decannulated.
Conclusions: This series reflects current trends in the indications for paediatric tracheostomy, with chronic lung disease of prematurity being the most common indication.
© 2015 Elsevier Ireland Ltd. All rights reserved.

1. Introduction

Tracheostomy is a life saving operation, however, it is associated with potential serious complications. Recent publications suggest that the indications for tracheostomy in children have been changing considerably over recent years, in comparison to the adult population where indications have remained the same. Life threatening infections, such as diphtheria, were historically the main indication for paediatric tracheostomy [1]. However with the introduction of modern immunizations, life-threatening infections are now less common, and subsequently the need for tracheostomy. Advances in neonatal medicine now allow children to stay on long-term ventilator support for many weeks to months. Tracheostomies help facilitate pulmonary toilet in such patients, and reduce chronic laryngotraceal lesions related to long term intubation, such as subglottic stenosis [2] Paediatric tracheostomy is technically more demanding than one performed on an adult, due to the size of the patient and the limited extent of the operating field. Paediatric patients have a higher reported mortality, morbidity and complication rate, particularly in pre-term infants, compared to the adult population [3,4]. The aim of this paper was to review our 11 year experience of tracheostomies and compare it to the airway workload in our department.

2. Materials and methods

A retrospective review of all tracheostomies performed at the Royal Hospital of Sick Children between January 2001 and 1st September 2012 was carried out. The Royal Hospital for Sick Children, Glasgow, serves a population of around 1 million people, as well as being the tertiary referral centre in Scotland for paediatric airway services. Patient case notes were analysed with respect to the following variables: age, demographics, indication for procedure, co-morbidities, length of stay, decannulation, trachea-cutaneous fistula closure, and complications experienced. ENT surgeons carry out all tracheostomies performed in the hospital. All tracheostomies were performed under general anaesthetic. A standard tracheostomy procedure is used, with a horizontal skin incision. Once trachea is visible, a vertical incision is made in the trachea, and two parallel stay sutures are placed on either side of the incision. Stomal maturation sutures are placed from trachea to skin, which became routine practice in 2010. The

* Corresponding author. Tel.: +0141 201 6940; fax: +0141 201 0865.
E-mail address: catriona.douglas2@nhs.net (C.M. Douglas).

http://dx.doi.org/10.1016/j.ijporl.2015.07.022
0165-5876/© 2015 Elsevier Ireland Ltd. All rights reserved.

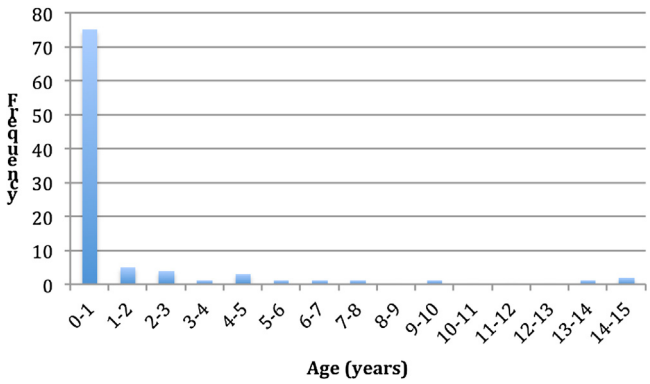

Fig. 1. Age distribution of patients.

tracheostomy tube is placed. All patients were admitted to the paediatric intensive care unit post operatively. Tracheostomy tubes were changed bedside 2–3 days after the operation.

3. Results

3.1. Patient demographics

One hundred and eleven patients underwent a tracheostomy over an 11-year period, with notes available for 95 patients. All tracheostomies were performed by ENT surgeons (consultants or surgeons in training with consultant supervision). There were 56 (59%) males and 39 (41%) females. Age at time of tracheostomy ranged from one day to 15 years, the mean age of tracheostomy insertion was 69 weeks. The majority of patients, 75 (79%), were under one year old when they had their tracheostomy performed, see Fig. 1, with the majority of these patients being under 5 months of age, see Fig. 2. The mean weight at time of tracheostomy was 6.9 kg (1.57–60 kg).

3.2. Preoperative ventilation status

Fifty two patients (55%) were intubated prior to the tracheostomy insertion, 43 patients (45%) were not intubated prior to tracheostomy insertion. The mean length of intubation prior to tracheostomy was 6 days (1–26). 84 (88%) patients had their procedure performed as an elective case, 11 (12%) had their tracheostomy performed as an emergency.

3.3. Indications

Indications for the tracheostomy insertion are shown in Table 1. The most common indication was long-term ventilation (20%), followed by craniofacial abnormality causing airway obstruction (18%), followed by subglottic stenosis (14%).

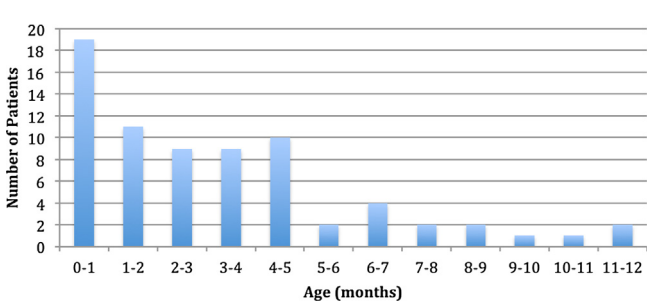

Fig. 2. Number of patients under 12 months of age.

Table 1
Indications for tracheostomy.

Indication	Number of patients (%)
Acute neurological insult	2 (2)
Laryngomalacia	2 (2)
Cardiac patient for weaning	2 (2)
Bilateral vocal cord palsy	10 (11)
Long term ventilation	30 (32)
• Tracheobronchomalacia	
• Chronic neurodevelopmental disorder	
• Congenital hypoventilation syndrome	
• Diaphragmatic hernia	
• Cardiac/right phrenic nerve palsy	
Chronic lung disease of prematurity	7 (7)
Recurrent difficult intubation	1 (1)
Chronic neurological disorder	2 (2)
Obstructive sleep apnoea	1 (1)
Craniofacial abnormality causing upper airway obstruction	17 (18)
Macroglossia causing airway obstruction	1 (1)
Neck mass causing airway obstruction	2 (2)
Subglottic stenosis	13 (14)
Other	5 (5)

3.4. Complications

Complications were divided into early post operative and late postoperative, see Table 2. Early postoperative complications included one pneumothorax, 5 patients with bleeding from their stoma and 5 episodes of tube displacement requiring emergency tube change. Only granulation tissue requiring excision was included as a postoperative complication, almost all patients developed mild stomal granulation which was treated with topical steroid cream and/or topical silver nitrate cautery.

3.5. Length of stay and decannulation

The mean length of stay for admission during which the tracheostomy was performed was 18 weeks (1–91). Thirty-five (37%) of patients went on to have decannulation of their tracheostomy tube. The mean duration of tracheostomy prior to decannulation was 28 months (1–132). Twenty-seven (77%) of the 35 decannulated patients underwent a routine microlaryngoscopy and bronchoscopy (MLB) prior to decannulation. 9 (33%) of the 27 patients who underwent MLB prior to decannulation were found to have suprastomal granulation tissue and 3 patients were found to have new subglottic stenosis.

3.6. Number of tracheostomies and MLBs

The mean number of tracheostomies performed each year is 9 (3–12), with the numbers remaining relatively stable over the study period, Fig. 3. In comparison, there has been a steady increased in the number of microlaryngoscopy and bronchoscopies done each year, Fig. 4.

Table 2
Tracheostomy complications.

Complication	Number of patients (%)
Early post operative	5 (5)
- Bleeding	1 (1)
- Pneumothorax	5 (5)
- Tube displacement	
Late post operative	13 (14)
- Wound infection	9 (9)
- Granulation tissue requiring intervention	3 (3)
- Subglottis stenosis	19 (20)
- Tracheocutaneous fistula	

Please cite this article in press as: C.M. Douglas, et al., Paediatric tracheostomy—An 11 year experience at a Scottish paediatric tertiary referral centre, Int. J. Pediatr. Otorhinolaryngol. (2015), http://dx.doi.org/10.1016/j.ijporl.2015.07.022

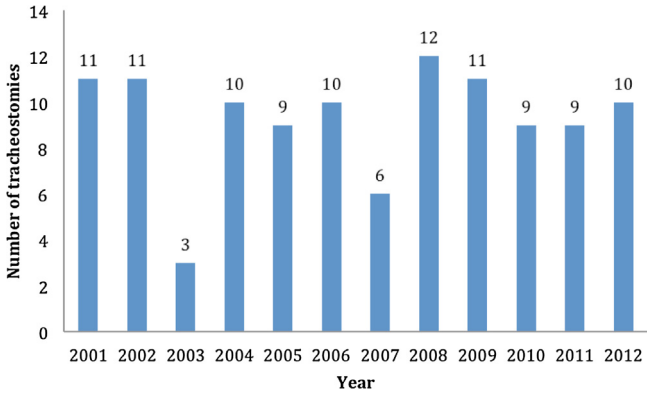

Fig. 3. Number of tracheostomies performed each year.

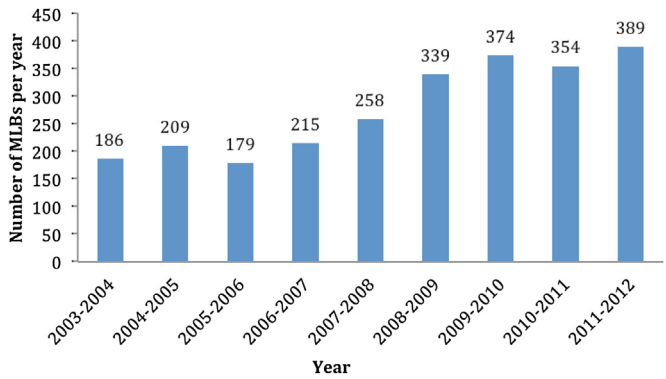

Fig. 4. Number of microlaryngoscopy and bronchoscopy procedures performed each year.

3.7. Mortality

A total of 21 (21%) patients have died. Seventeen (18%) of these patients have died as a result of disease related progress or complications, unrelated to tracheostomy, see Table 3. We were unable to identify the cause of death in four patients.

4. Discussion

Historically the most common indication for paediatric tracheostomy has been for the management of upper airway obstruction caused by infection [5]. However, no child in this series had a tracheostomy for infection of the upper airway, see Table 1. In this series of patients the most common indication was long-term ventilation (20%), followed by craniofacial abnormality causing airway obstruction (18%), followed by subglottic stenosis (14%). Several series have documented the changing indications for tracheostomy in children over the past decade [6,7]. In the developed countries acute epiglottitis and laryngotracheobronchitis no longer represent an indication for tracheostomy due to the introduction of the endotracheal tube and the haemophilus influenza type B vaccine for acute epiglottitis [8,9]. As discussed above, tracheostomy for long-term ventilation was the most common indication in our series, a result comparable with other large series from the United Kingdom. The Great Ormond Street Hospital and Alder Hay Hospital both reported long term ventilation as the most common indication for tracheostomies in their series [10,11]. Due to the improvement in neonatal intensive care, there has been an increased survival in these young neonates although this has subsequently resulted in more morbidity, mainly bronchopulmonary dysplasia, with a subsequent need for tracheostomy [12]. There has also been a trend towards patients with significant medically complex problems surviving long term, and these patients often require long-term ventilation via a tracheostomy, as highlighted in this series and comparable to other series [10]. Although well established that long-term ventilation is the most common indication for tracheostomy in the UK and many countries this is not the case in all countries. Ozmen et al. from Turkey reported that in their series of 282 patients, airway obstruction accounted for 72%, while tracheostomy for long-term ventilation only accounted for 28% [13]. A large series from Starship Hospital, New Zealand showed that upper airway obstruction was the most common indication, accounting for 70% of their patients [7]. However, despite the frequency of indications varying between studies, the general trend over the past few decades does appear to be that tracheostomies are now commonly being performed for chronic conditions [5,6,14,15].

There have been several studies reporting a general decrease in the number of paediatric tracheostomies performed over time [16,17]. It has been suggested that people are moving away from tracheostomies for short-term management of airway problems [13]. In our series, we saw the number of tracheostomies performed each year remaining relatively stable, with an average of 9 per year, see Fig. 2. This is comparable to the recent series published by Ogilvie et al. who report a plateau in tracheostomy numbers over the past ten years [6]. The Royal Hospital for Sick Children, Glasgow is the tertiary referral center for airway problems in Scotland. Interestingly, although the number of tracheostomies has remained stable over the past decade, the number of microlaryngoscopies and broncosopies has been increasing, see Fig. 3. It may be that more airway pathology is now being treated with endoscopic techniques, and conditions that historically would have been treated with a tracheostomy, such as severe laryngomalacia, are now being treated endoscopically.

Fifty five percent of patients in this series were intubated prior to tracheostomy insertion. There has been a shift in practice of intubation prior to tracheostomy over the past few decades. In the 1970s, intubation of greater than 8 days was generally the threshold, after which a tracheostomy was recommended [18]. However, with the advent of improved ITU care and improved endotracheal tubes the parameters are set according to the patient's clinical and endoscopic parameters. In our series, the average length of intubation prior to tracheostomy was 6 days, with a range of 1–26 days. This is slightly less than comparable series, with the average range prior to tracheostomy being between 2 and 134 days [10,14,19].

The morbidity and mortality of paediatric tracheostomy has been reported to be 2 to 3 times higher than in adult patients [5], however a recent publication has suggested that complications

Table 3
Cause of death.

Cause of Death	Number of patients (%)
Lung disease	4 (4)
Cardiac complications	4 (4)
Infected VP shunt	1 (1)
Brown-Vialetto-Van Laere Syndrome	1 (1)
Multi-organ failure	1 (1)
Neuromyotinia	1 (1)
Support withdrawn due to very poor quality of life.	3 (3)
Brain tumour recurrence	1 (1)
Complications of chronic disease	1 (1)
Unknown	4 (4)

are similar [20]. Major complications include haemorrhage, pneumothorax and accidental decannulation. Seventy-nine percent of patients in this series were under one year of age. Fifty-seven percent of patients developed some form of complication. Complications are known to be higher in this vulnerable younger age group [21]. Published rates of complication range from 5 to 49% in several case series, making our series comparable with others [5,18,19]. Complications seem to be related to younger patients, which may be related to the prolonged need for tracheostomy in this age group with co-morbid pathology along with the small size of the airway [22]. It may also be related to the smaller diameter of the trachea and its greater flexibility. The diameter of the trachea is approximately 6 mm by 1–4 years, 8 mm by 4–8 years of age and 10 mm by 8–10 years of age, so even a small reduction in diameter can potentially obstruct the airway [18]. It has also been noted that in patients with bronchopulmonary dysplasia, patients get more viscous and abundant secretions, thus increasing the risk of complications [23]. Twenty-one patients in this series had died, giving an overall mortality rate of 21%. Although we had four patients in whom we could not identify the cause of death, it was not thought that their death was directly related to their tracheostomy tube. The overall mortality rate in patients with a tracheostomy appears to be high. Corbett et al. reported an overall mortality of 19.6% comparable with our rate of 21% [10]. As highlighted earlier, in our series of patients, there was a complex patient mix with many patients that have long-term incurable conditions. Again, this is comparable to other reported series [5,16]. It is likely that the high mortality within paediatric tracheostomy series is related to the complex co-morbidities that many of these patients have. Although we didn't have any known deaths from acute tracheostomy obstruction or displacement, this will continue to be an ongoing hazard for such patients. This also highlights the need for comprehensive training in tracheostomy care for health care professionals involved in the care of such complex patients.

Decannulation was carried out successfully in 37% of our patients, comparable to Ozmen et al., who reported a decannulation rate of 35% and Carr et al. who reported a decannulation rate of 34% [13,21]. However, these rates highlight that the severity of the primary disease often precludes decannulation in most patients.

5. Conclusion

This series highlights the changing indications in paediatric tracheostomy. This is emphasized by the fact that only 37% of our series were successfully decannulated, highlighting again the change in paediatric tracheostomy over the past decade.

References

[1] A.C. Swift, J.H. Rogers, The changing indications for tracheostomy in children, J. Laryngol. Otol. 101 (12) (1987) 1258–1262.
[2] J.D. Carron, C.S. Derkay, G.L. Strope, J.E. Nosonchuk, D.H. Darrow, Pediatric tracheotomies: changing indications and outcomes, Laryngoscope 110 (7) (2000) 1099–1104.
[3] M.A. Kenna, J.S. Reilly, S.E. Stool, Tracheotomy in the preterm infant, Ann. Otol. Rhinol. Laryngol. 96 (Jan–Feb (1) Pt 1) (1987) 68–71.
[4] R.F. Wetmore, S.D. Handler, W.P. Potsic, Pediatric tracheostomy. Experience during the past decade, Ann. Otol. Rhinol. Laryngol. 91 (Nov–Dec (6) Pt 1) (1982) 628–632.
[5] B. Kremer, A.I. Botos-Kremer, H.E. Eckel, G. Schlondorff, Indications, complications, and surgical techniques for pediatric tracheostomies—an update, J. Pediatr. Surg. 37 (11) (2002) 1556–1562.
[6] L.N. Ogilvie, J.K. Kozak, S. Chiu, R.J. Adderley, F.K. Kozak, Changes in pediatric tracheostomy 1982–2011: a Canadian tertiary children's hospital review, J. Pediatr. Surg. 49 (11) (2014) 1549–1553.
[7] M. Mahadevan, C. Barber, L. Salkeld, G. Douglas, N. Mills, Pediatric tracheotomy: 17 year review, Int. J. Pediatr. Otorhinolaryngol. 71 (12) (2007) 1829–1835.
[8] P.M. Palmer, J.M. Dutton, T.M. McCulloch, R.J. Smith, Trends in the use of tracheotomy in the pediatric patient: the Iowa experience, Head Neck 17 (Jul–Aug (4)) (1995) 328–333.
[9] M.J. Donnelly, P.D. Lacey, A.J. Maguire, A twenty year (1971–1990) review of tracheostomies in a major paediatric hospital, Int. J. Pediatr. Otorhinolaryngol. 35 (Mar (1)) (1996) 1–9.
[10] H.J. Corbett, K.S. Mann, I. Mitra, E.C. Jesudason, P.D. Losty, R.W. Clarke, Tracheostomy—a 10-year experience from a UK pediatric surgical center, J. Pediatr. Surg. 42 (7) (2007) 1251–1254.
[11] P.J. Hadfield, R.V. Lloyd-Faulconbridge, J. Almeyda, D.M. Albert, C.M. Bailey, The changing indications for paediatric tracheostomy, Int. J. Pediatr. Otorhinolaryngol. 67 (1) (2003) 7–10.
[12] G.M. Stoelhorst, M. Rijken, S.E. Martens, et al., Changes in neonatology: comparison of two cohorts of very preterm infants (gestational age <32 weeks): the Project On Preterm and Small for Gestational Age Infants 1983 and the Leiden Follow-Up Project on Prematurity 1996–1997, Pediatrics 115 (2) (2005) 396–405.
[13] S. Ozmen, O.A. Ozmen, O.F. Unal, Pediatric tracheotomies: a 37-year experience in 282 children, Int. J. Pediatr. Otorhinolaryngol. 73 (7) (2009) 959–961.
[14] J.J. Song, I.J. Choi, H. Chang, et al., Pediatric tracheostomy revisited: a nine-year experience using horizontal intercartilaginous incision, Laryngoscope 125 (2) (2015) 485–492.
[15] J.L. Funamura, B. Durbin-Johnson, T.T. Tollefson, J. Harrison, C.W. Senders, Pediatric tracheotomy: indications and decannulation outcomes, Laryngoscope 124 (8) (2014) 1952–1958.
[16] R.F. Wetmore, R.R. Marsh, M.E. Thompson, L.W. Tom, Pediatric tracheostomy: a changing procedure? Ann. Otol. Rhinol. Laryngol. 108 (7 Pt 1) (1999) 695–699.
[17] P. Arcand, J. Granger, Pediatric tracheostomies: changing trends, J. Otolaryngol. 17 (Apr (2)) (1988) 121–124.
[18] A. Serra, S. Cocuzza, M.R. Longo, C. Grillo, M. Bonfiglio, P. Pavone, Tracheostomy in childhood: new causes for an old strategy, Eur. Rev. Med. Pharmacol. Sci. 16 (Nov (12)) (2012) 1719–1722.
[19] K.I. Midwinter, D. Hodgson, M. Yardley, Paediatric epiglottitis: the influence of the Haemophilus influenzae b vaccine, a ten-year review in the Sheffield region, Clin. Otolaryngol. Allied Sci. 24 (Sep (5)) (1999) 447–448.
[20] N. Suslu, G. Ermutlu, U. Akyol, Pediatric tracheotomy: comparison of indications and complications between children and adults, Turk. J. Pediatr. 54 (Sep–Oct (5)) (2012) 497–501.
[21] M.M. Carr, C.P. Poje, L. Kingston, D. Kielma, C. Heard, Complications in pediatric tracheostomies, Laryngoscope 111 (11 Pt 1) (2001) 1925–1928.
[22] G.J. Gianoli, R.H. Miller, J.L. Guarisco, Tracheotomy in the first year of life, Ann. Otol. Rhinol. Laryngol. 99 (11) (1990) 896–901.
[23] J.S. Schlessel, R.G. Harper, H. Rappa, K. Kenigsberg, S. Khanna, Tracheostomy: acute and long-term mortality and morbidity in very low birth weight premature infants, J. Pediatr. Surg. 28 (7) (1993) 873–876.

Please cite this article in press as: C.M. Douglas, et al., Paediatric tracheostomy—An 11 year experience at a Scottish paediatric tertiary referral centre, Int. J. Pediatr. Otorhinolaryngol. (2015), http://dx.doi.org/10.1016/j.ijporl.2015.07.022